Laboratory job 19

Motor starting using wye-delta switch

One method of reducing the starting current of three phase motors is using wye-delta switch. This method is used for three phase motors designed to run delta connected (the most common connection) which also means that the motors rating is 380V. Using this method the motor is started with a voltage per phase reduced to 58%(220/380x100) and correspondingly the starting current is low.

1. How many leads must come out from the motor so that they can be connected wye when starting and then delta when running the motor?
2. Explain operation of the circuit and construct it. Have the circuit checked by the instructor before you switch on the power.
3. Read the power rating of the motor from its name plate and then:
    a) Select the rating of wye-delta switch required.
    b) Choose the rating of fuses for fault protection
    c) Select the rating of the overload relay for overload protection
5. Explain using a diagram how a wye-delta switch reduces the starting current of motor.

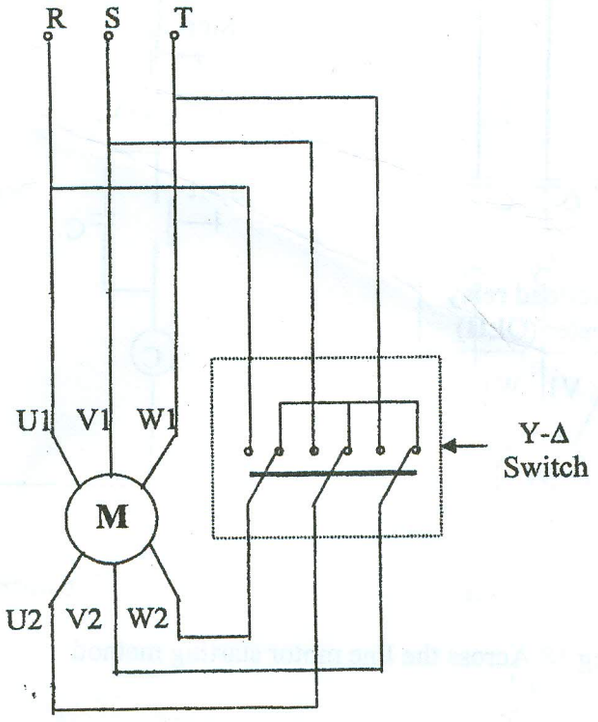


Fig 19 Y-Δ motor starting circuit using Y-Δ switch